



Tire Highway Speed Uniformity

1 Scope

Note: Nothing in this standard supersedes applicable laws and regulations unless specific exemption has been obtained.

Note: In the event of conflict between the English and domestic language, the English language shall take precedence.

1.1 Purpose. This document provides details for testing passenger car and light truck tires for highway speed uniformity (HSU).

1.2 Foreword. HSU data is used in such applications as vehicle sensitivity tests, auditing assembly plant tire-wheel assemblies, analyzing tires-wheel assemblies from complaint vehicles, and auditing tire production variation. The HSU parameters of interest include primarily the first 4 harmonics, magnitudes and phases.

1.3 Applicability. This procedure applies to the testing of tires on an HSU machine at zero slip angle and zero inclination angle.

2 References

Note: Only the latest approved standards are applicable unless otherwise specified.

2.1 External Standards/Specifications.

SAE J332

2.2 GM Standards/Specifications.

None

3 Resources

3.1 Facilities. Not applicable.

3.2 Equipment. All details associated with the test apparatus are included in SAE J332.

3.2.1 The roadwheel should be covered with either medium grit 3M Safety Walk or 80 grit 3M ite. The surface should be replaced at routine intervals to eliminate it as a source of testing variability.

3.3 Test Vehicle/Test Piece. A tire-wheel assembly is required to run an HSU test. The wheel is either the wheel from the assembly as it came from a complaint vehicle, or in the case of an audit of multiple tires, a set of production or test

wheels meeting the radial and axial runout requirements of SAE J332. The test requester will specify the width and rim flange of the wheel.

3.4 Test Time.

Calendar time: 1 day

Test hours: 1 hour

Coordination hours: 1 hour

3.5 Test Required Information. Required information includes tire size, construction #, load rating, service description, and wheel width. Additional information required may vary by test site.

3.6 Personnel/Skills. Test operator should have adequate training and be familiar with the local operator's procedure.

4 Procedure.

4.1 Preparation.

4.1.1 Tires must be mounted on the appropriate wheels (see paragraph 3.3).

4.1.2 All tires must be mounted in the proper orientation with respect to the front face of the test wheels. Whitewall tires must be mounted whitewall out with respect to the front face of the wheels. Blackwall tires must be mounted with the cosmetic or intended outboard sidewall mounted out with respect to the front face of the test wheels.

4.1.3 Assemblies or individual tires and wheels should be at ambient room temperature ($24.0 \pm 3.0^{\circ}\text{C}$) for a minimum of 2 h before an HSU test.

4.1.4 The assembly should always be mounted on the machine in the standard position, in which the valve stem is aligned with the 0 mark on the spindle. If this is not possible because the spindle studs do not line up with the boltholes, the assembly is rotated clockwise until the studs and boltholes do line up. The angle through which the assembly was rotated is then used to correct the phase angles of the Fast Fourier Transform (FFT).

4.1.5 Minimizing the radial runout is very important to minimize mounting contribution to uniformity. Follow local mounting instructions using expandable collets, and other mounting hardware

in order to optimize the centering of the machine mounting.

4.2 Conditions.

4.2.1 Environmental Conditions. Ambient temperature must be kept at $24.0 \pm 3.0^{\circ}\text{C}$. A method for monitoring ambient temperature must be in place.

4.2.2 Test Conditions.

4.2.2.1 Load and Pressure.

Tires must be tested at standard test conditions unless otherwise specified by the tire test request. Standard test conditions are defined in Table 1.

Table 1: Calculation of Standard HSU Test Conditions

Tire Type	Load Range	Test Load	Test Pressure
Passenger Car	all	70% of max load	210 kPa
Light Truck or Commercial	all	70% of load capacity at 350 kPa	350 kPa

4.2.2.2 Speed. The typical test speeds used to evaluate HSU is a range of 8 speeds from 80 to 150 kph in 10 kph increments, in both clockwise and counter-clockwise directions. However, a single speed, 120 kph, may be used to satisfy certain production audit requirement. Other speeds, not listed here, may also be requested depending upon the specific data requirements.

4.2.3 Deviations from the requirements of the procedure shall have been agreed upon. Such requirements shall be specified on component drawings, test certificates, reports, etc.

4.3 Instructions.

4.3.1 Highway Speed Uniformity

4.3.1.1 Warm-Up: The first step is to warm up the tire. The tire warm-up is performed at the test load and pressure and 120 kph in the clockwise direction for 20 m. If a tire is put through multiple tests that occur within 5 m of each other, only the first test of the tire requires a warm-up.

4.3.1.2. Data Collection: The set points (load and speed combinations) are run in the following order.

4.3.1.2.1 The roadwheel is first running in the clockwise direction.

4.3.1.2.2 The speed of the first test condition is set.

4.3.1.2.3 The load of the first test condition is set.

4.3.1.2.4 The composite data is then acquired once the tire has reached a steady state condition. 8 revolutions of data is acquired.

4.3.1.2.5 Repeat paragraphs 4.3.1.2.2 to 4.3.1.2.4 for all clockwise speed conditions in descending order.

4.3.1.2.6 The unbalanced data for the clockwise (CW) direction is now collected using the following paragraphs:

4.2.1.2.6.1 Accelerate the roadwheel to a speed incrementally higher than the highest test speed.

4.2.1.2.6.2 Move the test wheel away from the roadwheel so that it is free spinning.

4.2.1.2.6.3 As the tire is decelerating, collect unbalanced data at each of the test speed conditions. 8 revolutions of data is acquired for the unbalanced data at each test speed condition.

4.2.1.2.6.4 The tire is reloaded to the roadwheel at the last test speed. Caution should be taken not to reload tire when large speed differences exist.

4.3.1.2.7 Change direction of the roadwheel to counter clockwise (CCW) and repeat paragraphs 4.3.1.2.2 to 4.3.1.2.6 for all speed conditions in ascending order.

4.3.1.3 Analysis: Detailed Analysis is described in paragraph 5.1. In summary, the data is analyzed in three ways using FFT. The composite data is analyzed by itself, the unbalance data is analyzed by itself, and the difference of the composite and unbalance, called the assembly data, is analyzed. The machine contribution can be subtracted out of both the composite and assembly data if desired.

4.3.2 Machine Contribution

4.3.2.1 Machine contributions can be calculated for every harmonic of each parameter at any speed and load. However, typically only radial first harmonic (R1H) and fore-aft first harmonic (F1H) are subtracted out of the composite and assembly data. The following steps detail the procedure.

4.3.2.2 Run a standard warm-up for the assembly in its standard position on the machine. See paragraph 4.3.1.1 above for the warm-up procedure.

4.3.2.3 Acquire the composite data at all necessary loads and speeds (paragraph 4.3.1.2 above). It is not necessary to acquire the unbalance data.

4.3.2.4 Remount the assembly on the machine, rotating the assembly 1 bolthole clockwise relative to the standard position on the spindle.

4.3.2.5 Repeat paragraph 4.3.2.3 above.

4.3.2.6 Repeat paragraph 4.3.2.3 and 4.3.2.4 for all other boltholes of the assembly.

4.3.2.7 In the cases of machines where the wheel is not dismounted from the machine to mount the tire, the above process would be done by rotating the tire with respect to the wheel in a set of equal angle rotations, making up one full rotation.

5 Data

5.1 Calculations.

5.1.1 Highway Speed Uniformity

5.1.1.1 All 8 revolutions of the acquired data are averaged: $F_x(i)$, $F_y(i)$, $F_z(i)$, $M_x(i)$ and $M_z(i)$.

5.1.1.2 The magnitudes and phase angles of radial, fore/aft and lateral forces, as well as overturning moment and aligning torque for each harmonic are calculated based on the following formulations.

$$F_z(i) = \sum_{j=1}^n R_{jH} * \cos\left(\frac{2 * \pi}{j * T} * \frac{i}{m} + R_{jP}\right)$$

Where:

$F_z(i)$ is measured as an 8-rev-averaged radial force in Society of Automotive Engineer (SAE) coordinate system at each sample points in one revolution

i is variable index of sample points $i=1, 2, \dots, m$

m is number of sample points per revolution, such as 64 256

j is variable index of harmonic $j=1, 2, \dots, n$ n is number of harmonic, such as 16, 32

R_{jH} is decomposed magnitude (Newtons) of radial force at harmonic j^{TH} or frequency of $1/(j*T)$

R_{jP} is decomposed phase angle (Deg) of radial force at harmonic j^{TH} or frequency of $1/(j*T)$

T is time taken for tire to rotate one cycle at specific speed and load

$$F_x(i) = \sum_{j=1}^n F_{jH} * \cos\left(\frac{2 * \pi}{j * T} * \frac{i}{m} + F_{jP}\right)$$

Where:

$F_x(i)$ is measured as an 8-rev-averaged fore/aft force in SAE coordinate system at each sample points in one revolution

F_{jH} is decomposed magnitude (Newtons) of fore/aft force at harmonic j^{TH} or frequency of $1/(j*T)$

F_{jP} is decomposed phase angle (Deg) of fore/aft force at harmonic j^{TH} or frequency of $1/(j*T)$

$$F_y(i) = \sum_{j=1}^n L_{jH} * \cos\left(\frac{2 * \pi}{j * T} * \frac{i}{m} + L_{jP}\right)$$

Where:

$F_y(i)$ is measured as an 8-rev-averaged lateral force in SAE coordinate system at each sample points in one revolution

L_{jH} is decomposed magnitude (Newtons) of lateral force at harmonic j^{TH} or frequency of $1/(j*T)$

L_{jP} is decomposed phase angle (Deg) of lateral force at harmonic j^{TH} or frequency of $1/(j*T)$

$$M_x(i) = \sum_{j=1}^n M_{xjH} * \cos\left(\frac{2 * \pi}{j * T} * \frac{i}{m} + M_{xjP}\right)$$

Where:

$M_x(i)$ is measured as an 8-rev-averaged overturning moment in SAE coordinate system at each sample points in one revolution

M_{xjH} is decomposed magnitude (Newton-Meter) of overturning moment at harmonic j^{TH} or frequency of $1/(j*T)$

M_{xjP} is decomposed phase angle (Deg) of overturning moment at harmonic j^{TH} or frequency of $1/(j*T)$

$$M_z(i) = \sum_{j=1}^n M_{zjH} * \cos\left(\frac{2 * \pi}{j * T} * \frac{i}{m} + M_{zjP}\right)$$

Where:

$M_z(i)$ is measured as an 8-rev-averaged aligning torque in SAE coordinate system at each sample points in one revolution

M_{zjH} is decomposed magnitude (Newton-Meter) of aligning torque at harmonic j^{TH} or frequency of $1/(j*T)$

M_{zjP} is decomposed phase angle (Deg) of aligning torque at harmonic j^{TH} or frequency of $1/(j*T)$

5.1.1.3 The FFT is run on 3 data sets:

- Composite data.
- Unbalance data.
- Assembly data (composite minus unbalance).

5.1.1.4 Tire phase and magnitudes for the first 16 harmonics are calculated. Typically the first four harmonics are of primary interest and are summarized in the report. Calculated harmonics should not exceed half the pulse count of the

encoder. Also, reported harmonics should not exceed the capability of the machines in terms of machine natural frequency's influence on accuracy of the data.

5.1.1.5 Phase angle is always reported as a physical spot on the tire measured clockwise from the 0 mark of the spindle. The phase angle is always positive, so the n^{th} harmonic phase angle can only be from 0° to $360^\circ/n$.

5.1.2 Machine Contribution

5.1.2.1 Machine contributions may be quantified periodically. There are two primary uses for this information; 1. as a check to quantify the machine effect on HSU data compared to historical levels, or 2. to remove the machine effects from a specific data set such as a large sample of audit tires. In the last case, the machine contribution can be removed from both the composite and assembly data sets through vector subtraction.

5.1.2.2 Machine contributions can be easily calculated provided the angle of rotations are all equal in magnitude, and that a complete set of orientations have been measured. The machine contribution is then the average of the vectors for a given parameter and harmonic.

5.1.2.3 In general, machine contributions are expected to be relatively small with respect to the magnitudes of HSU being measured. If this is not true, then effort should be focused on reducing the magnitude of the machine contributions through root cause analysis of the HSU measurement system.

5.1.2.4 Through historical monitoring of machine contributions, one can discern the acceptability of the current equipment state and potentially identify maintenance issues.

5.2 Interpretation of Results. The requester of the test will review the measured data, according to their specific needs.

5.3 Test Documentation. For test improvement process (TIP) approved data being uploaded to the GM Global Tire Database, follow the electronic file formats in Appendix A. Otherwise, document the test results as shown in the sample test report in Appendix B.

6 Safety

This standard may involve hazardous materials, operations, and equipment. This standard does not propose to address all the safety problems associated with its use. It is the responsibility of the user of the method to establish appropriate safety and health practices and determine the applicability of regulatory limitations prior to use.

6.1 Proper procedures and safety equipment must be in place to ensure operator safety when installing test tires, operating the test machine, and removing test tires.

7 Notes

7.1 Glossary.

Applied Load: The force existing between the tire and the roadwheel during an HSU test.

Assembly Data: The difference between composite and unbalance data. Assembly data is calculated as composite minus unbalance.

Composite Data: The data acquired while the tire is run in a steady-state condition loaded against the roadwheel.

Composite Magnitude: The peak-to-peak magnitude of the force variation over one complete tire rotation, 360 degrees.

First Harmonic: The peak-to-peak magnitudes of the lowest or fundamental frequency of the force variation signal over one complete tire rotation, 360 degrees.

FxH: Fore/Aft Force Variation Vector, magnitude and phase, for the x^{th} harmonic. The fore/aft direction is also referred to as Fx in SAE coordinates.

Inflation Pressure: The capped tire inflation pressure during an HSU test.

LxH: Lateral Force Variation Vector, magnitude and phase, for the x^{th} harmonic. The lateral direction is also referred to as Fy in SAE coordinates.

Machine Contribution: A measure of the machine's contribution to the non uniformity that appears in the test data. Machine contributions can come from an out of balance spindle and eccentricities of the assembly when it is mounted to the machine. They are typically first harmonic phenomena, but they can appear in other harmonics. Machine contributions are different for every colleting/wheel construction combination.

Magnitude: The length of the vector that describes the force or moment of a harmonic of the tire. Force magnitudes are reported in Newtons. Moment magnitudes are reported in Newton meters. Magnitudes are always positive.

MXxH: Overturning Moment Variation Vector, magnitude and phase, for the x^{th} harmonic. The overturning moment direction is also referred to as Mx in SAE coordinates.

MZxH: Aligning Torque Variation Vector, magnitude and phase, for the x^{th} harmonic. The

aligning torque direction is also referred to as M_z in SAE coordinates.

n^{th} Harmonic: The peak-to-peak magnitude of the frequency that is n times the lowest or fundamental frequency of the force variation signal over one complete tire rotation, 360 degrees.

Phase Angle: The angle of the vector that describes a harmonic of the tire. By GM standards, phase is reported in degrees as a physical spot on the tire, measured clockwise from the valve stem of the assembly. Phases are always positive. Thus, the first harmonic phase will always be from 0° to 360° . The second harmonic phase will always be from 0° to 180° . The n^{th} harmonic phase will always be from 0° to $360^\circ/n$. The 0° reference marks the valve stem of the assembly when it is on a line segment connecting the center of the roadwheel to the center of the assembly, as shown below.

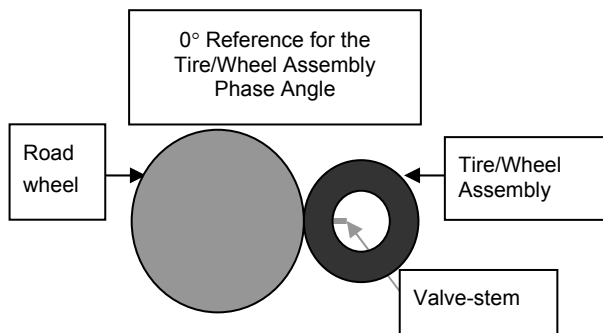


Figure 1: Phase Angle Reference

R_xH: Radial Force Variation Vector, magnitude and phase, for the x^{th} harmonic. The radial direction is also referred to as F_z in SAE coordinates.

Speed: The speed of the tire during data acquisition. Speed is selectable for a test.

Standard Position: The assembly is mounted on the machine such that the valve stem is aligned

with the 0 mark on the spindle. If this is not possible because studs on the spindle do not match up with the bolt holes of the assembly, the assembly is rotated clockwise until the studs and bolt holes are aligned. The angle of rotation is noted so that it can be subtracted out of the phase angles from the FFT's. This is the mounting position for all HSU tests unless specified otherwise.

Unbalance Data: The data acquired while the tire is free spinning, i.e. not touching the roadwheel.

Warm-up Speed: The speed of the tire during the warm-up before an HSU test.

7.2 Acronyms, Abbreviations, and Symbols.

CW	Clockwise
CCW	Counter Clockwise
FFT	Fast Fourier Transform
F1H	Fore-aft First Harmonic
HSU	Highway Speed Uniformity
R1H	Radial First Harmonic
SAE	Society of Automotive Engineer
TIP	Test Improvement Process

8 Coding System

This standard shall be referenced in other documents, drawings, VTS, CTS, etc. as follows:
Test to GMW15120

9 Release and Revisions

9.1 Release. This standard originated in February 2006, replacing T01HSU1. It was first approved by the Tire Global Subsystem Leadership Team in July 2006. It was first published in December 2006.

Appendix A:

Electronic File Format Definitions for GM Global Tire Database Submission

.TPC and .DAT File Naming Convention – HSU			
File Type	File Name	Explanation	Example
TPC	XXVMDDSS.TPC	XX – Supplier Code V – HSU M – Month (1-9 and A,B,C) DD – Day of Month SS – Test Number (ex. 01)	XXVB0426.TPC HSU Test (Third Character = V) Tested on November 4 ("B" = November, "04" = Test Day) Test Number 26

TPC File Format:

LINE#	VARIABLE	SIZE	EXAMPLE	COMMENTS
GENERAL TEST INFORMATION				
1	TPC File Name (*)	A14	TWVA0501.TPC	Name of this file (see naming convention)
2	Data File Name (*)	A14	TWVA0501.dat	
3	Report image File Name (*)	A14	TWVA0501.pdf	Not required
4	Test Number	A5	00005	Sequential numbers starting at 00001.
5	Tire Test Request No.	A8	00-421	Supplied by GM
6	Test Site	A3	xxx	Valid codes only (see GM Global Tire Database)
7	Tester	A3	xxx	Valid codes only (see GM Global Tire Database)
8	End Test Date	A10	07/19/2000	Month/Day/Year of final test point completion
9	Purpose/Comment	A50	Development	
10	Test Machine	A8	Dyno A	
11	Test Station	A8	Sta. 2	
12	Drum Diameter	F5.3	1.710	[m] Total OD including surface treatment thickness
13	Drum Surface	A12	3M 80 Grit	Description of test drum surface treatment
TIRE TEST INFORMATION				
14				Blank Line
15	Tire Supplier	A3	xxx	Valid codes only (see GM Global Tire Database)
16	Brand Name	A20	Steeltex R4S	Name of tire model
17	Size	A13	P235/60R16	
18	Load Rating	A2	SL	Valid codes only (see GM Global Tire Database)
19	Service Description	A4	110Q	Load Index and Speed Rating ('Single' category for Light Truck tires)
20	Serv. Desc. (LT Dual)	A4	108Q	Load Index and Speed Rating of 'Dual' category for Light Truck tires, if applicable
21	Sidewall Type	A5	BSW	Short description of sidewall type
22	Construction No.	A12	123456	
23	Tire ID	A14	105	Character string to identify tire

24	DOT Serial	A6	xx1797	
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Electronic File Format Definitions for GM Global Tire Database Submission

25	Wheel Type	A9	OE	OE or Precision
26	Rim Width	F3.1	7.0	In [in] , to define rim width
27				Blank Line
28	Tested at the SRC ?	A3	Yes	Yes or No. Is this test done at the standard reference load and pressure?
29				Blank Line
30				Blank Line
31	Wheel offset	F6.2	23.25	[mm] as necessary to define test requirement
32	Adapter Linear Offset	F6.2	23.25	[mm] as necessary to define test requirement
33	Adapter Phase Offset	I3	34	[deg] as necessary to define test requirement
34	Test Type	A3	HSU	"HSU" for HSU Test
TEST DATA				
35	# Test steps	I2	3	Number of sets of data information for load/speed/inflation. One set is stored for each Composite set point type in the INP file.
36				Blank Line
37				Blank Line
38				Blank Line
39	# Harmonic Values	I2	4	Number of Harmonic values written to this file for each data point.
The fields below are written to the file one field per line.				
40	Date	A10	07/19/2000	Month/Day/Year of test point completion
41	Time	A8	15:13:55	Hour:Minute:Second of last test point completion (24 hr. format)
42	Radial Load	I5	2148	[N] Test load
43	Inflation Pressure	I3	250	[kPa] Test inflation pressure
44	Speed	I3	120	[kph] Road Wheel Speed
45	Rotation Direction	A3	CW	CW or CCW
46	Label	A80		"H_Index FjH(N) LjH(N) RjH(N) MXjH(NM) MZjH(NM)"
47	Index FjH LjH RjH MXjH MyjH MZjH	I2 F6.1 F6.1 F6.1 F6.1 F6.1 F6.1	11 479.4 479.4 479.4 479.4 479.4 479.4	ASSEMBLY DATA ONLY Harmonic index I=1,2,3 4.... Fore/aft forces(N) for harmonic j per direction CW(CCW). Lateral forces(N) for harmonic j per direction CW(CCW). Radial forces(N) for harmonic j per direction CW(CCW). Overturning moment (N*M) for harmonic j per direction CW(CCW). Aligning torque (N*M) for harmonic j per direction CW(CCW).

Electronic File Format Definitions for GM Global Tire Database Submission

Field 47 is repeated for the number of harmonic values specified in field 39 . .				
This example reflects four reported harmonics. If a greater number is reported the line numbers below would change accordingly.				
51	Label	A80		"H_Index FjP(deg) LjP(deg) RjP(deg) MXjP(deg) MZjP(deg)"
52	Index	I2	12	Harmonic Index =1,2,3,4....
	FjP	I4	209	Phases (deg) of fore/aft forces for harmonic j per direction CW(CCW).
	LjP	I4	209	Phases (deg) of lateral forces for harmonic j per direction CW(CCW).
	RjP	I4	209	Phases (deg) of radial forces for harmonic j per direction CW(CCW).
	MXjP	I4	209	Phases (deg) of overturning moments for harmonic j per direction CW(CCW).
	MyjP	I4	209	
	MZjP	I4	209	Phases (deg) of aligning torques for harmonic j per direction CW(CCW).
Field 52 is repeated for the number of harmonic values specified in field 39 . .				
Repeat lines 40 to 55 for the remaining Data Points. The number of data points is defined in field 33..				

NOTE:

- The file name must reflect the final test step date of completion of that tire.
Unless specified otherwise, all values in fields are to be left-justified and the remaining field padded with spaces i.e., "00-421 ", not "00-421".
- Unless specified otherwise, each field is written to one line in the file.
- Empty line is to keep the same format as one in database interface.

Appendix B

Sample Test Report

Page 1 and Page 2 of the HSU Report below, illustrate the report format.

Page 1 reports the magnitudes of Fx and Fy

Page 2 reports the phase angles of Fx and Fy

The same formats can be used to report magnitudes and phases of the moments as required with a page 3 and page 4..

HIGHWAY SPEED UNIFORMITY RESULTS

Page 1 - Forces - Magnitude

Brand/Model Name: Brand / Model Name
Size/LR/Serv Descr: P245/50R17 SL 98V
Construction: xxxxxx
Rim Width: 7.0
Purpose: DEV

Work Request: 06-XXX
Test Load: 5880 N
Test Pressure: 210 kPa
Run at SRC (Y/N): Yes
Test Comp Date: 9/8/06

Test Site/Tester: TWS / TWS
Machine/Station: Dyno A / STA 2
Drum Diameter: 1.7
Drum Surface: 3M 80 Grit

Test #	Tire ID	DOT	Load	Speed	Clockwise (Right Side of Vehicle)								Counter-Clockwise (Left Side of Vehicle)							
					F1H	F2H	F3H	F4H	R1H	R2H	R3H	R4H	F1H	F2H	F3H	F4H	R1H	R2H	R3H	R4H
TWV90802	330RF	xxxxxx	5901	80	55	48	15	20	127	30	34	13	54	42	20	35	122	27	31	23
TWV90802	330RF	xxxxxx	5875	90	75	64	11	19	131	33	39	20	75	48	18	34	127	35	37	28
TWV90802	330RF	xxxxxx	5872	100	100	84	23	18	135	30	44	28	99	62	16	41	129	33	44	30
TWV90802	330RF	xxxxxx	5875	110	126	88	28	15	138	38	51	47	126	73	19	40	137	30	51	53
TWV90802	330RF	xxxxxx	5868	120	156	89	33	20	143	35	59	67	159	64	25	61	139	32	62	84
TWV90802	330RF	xxxxxx	5868	130	192	88	44	68	147	34	72	109	191	56	35	19	150	37	74	133
TWV90802	330RF	xxxxxx	5863	140	231	92	56	74	153	38	86	88	231	49	43	45	154	41	96	94
TWV90802	330RF	xxxxxx	5864	150	277	89	77	50	162	47	109	60	267	43	56	67	163	40	123	67
TWV90801	330LF	xxxxxx	5891	80	21	27	47	32	113	1	24	10	24	36	24	8	119	4	26	7
TWV90801	330LF	xxxxxx	5891	90	31	46	48	34	119	5	29	5	36	53	28	8	120	5	28	5
TWV90801	330LF	xxxxxx	5886	100	43	50	52	39	116	7	34	14	49	62	32	8	121	7	34	10
TWV90801	330LF	xxxxxx	5892	110	61	49	60	45	124	8	40	29	66	66	37	14	125	7	38	17
TWV90801	330LF	xxxxxx	5904	120	80	43	72	52	128	9	44	54	86	65	44	21	133	12	43	52
TWV90801	330LF	xxxxxx	5882	130	105	28	91	83	134	8	54	67	109	60	53	23	137	11	47	105
TWV90801	330LF	xxxxxx	5883	140	136	21	116	74	142	13	63	57	136	60	56	32	145	14	59	81
TWV90801	330LF	xxxxxx	5801	150	168	27	132	44	142	18	66	36	165	57	70	34	150	20	65	63
TWV90704	343RF	xxxxxx	5891	80	25	38	23	37	68	10	40	14	26	41	10	20	65	17	39	11
TWV90704	343RF	xxxxxx	5891	90	35	58	30	42	75	15	47	9	37	64	6	19	73	10	44	14
TWV90704	343RF	xxxxxx	5886	100	48	75	38	51	74	18	52	17	50	73	11	20	71	19	50	16
TWV90704	343RF	xxxxxx	5892	110	62	85	43	57	86	19	64	26	65	88	14	14	78	16	59	28
TWV90704	343RF	xxxxxx	5904	120	82	84	55	57	82	21	76	27	80	86	21	35	79	26	68	48
TWV90704	343RF	xxxxxx	5882	130	100	83	70	107	84	25	93	50	101	90	27	42	80	27	82	87
TWV90704	343RF	xxxxxx	5883	140	118	82	102	44	88	26	116	22	123	86	39	23	86	28	98	66
TWV90704	343RF	xxxxxx	5801	150	132	84	140	24	95	26	136	12	141	90	49	31	93	30	123	51
TWV90703	343LF	xxxxxx	5891	80	32	63	14	41	69	7	25	3	32	80	19	39	59	12	24	11
TWV90703	343LF	xxxxxx	5891	90	46	103	19	44	69	2	32	11	45	104	17	37	67	8	31	17
TWV90703	343LF	xxxxxx	5886	100	60	115	28	45	78	6	34	9	58	126	9	37	70	11	33	15
TWV90703	343LF	xxxxxx	5892	110	77	132	29	53	85	5	40	21	75	141	4	56	78	5	39	29
TWV90703	343LF	xxxxxx	5904	120	100	132	39	48	91	5	48	44	98	135	2	87	90	8	43	78
TWV90703	343LF	xxxxxx	5882	130	125	131	50	56	100	4	56	102	124	129	5	58	95	9	54	138
TWV90703	343LF	xxxxxx	5883	140	150	126	78	9	106	6	75	79	149	127	8	21	106	10	62	93
TWV90703	343LF	xxxxxx	5801	150	178	125	120	12	114	8	89	50	178	124	28	40	115	18	73	70

Sample Test Report

HIGHWAY SPEED UNIFORMITY RESULTS

Page 2 - Forces - Phase

Brand/Model Name: Brand / Model Name
 Size/LR/Serv Descr: P245/50R17 SL 98V
 Construction: xxxxxx
 Rim Width: 7.0
 Purpose: DEV

Work Request: 06-XXX
 Test Load: 5580 N
 Test Pressure: 210 kPa
 Run at SRC (Y/N): Yes
 Test Comp Date: 9/8/06

Test Site/Tester: TWS / TWS
 Machine/Station: Dyno A / STA 2
 Drum Diameter: 1.7
 Drum Surface: 3M 80 Grit

Test #	Tire ID	DOT	Load	Speed	Clockwise (Right Side of Vehicle)								Counter-Clockwise (Left Side of Vehicle)							
					F1P	F2P	F3P	F4P	R1P	R2P	R3P	R4P	F1P	F2P	F3P	F4P	R1P	R2P	R3P	R4P
TWV90802	330RF	xxxxxx	5901	80	138	87	62	3	138	87	62	3	138	87	62	3	138	87	62	3
TWV90802	330RF	xxxxxx	5875	90	134	84	72	0	134	84	72	0	134	84	72	0	134	84	72	0
TWV90802	330RF	xxxxxx	5872	100	131	75	73	88	131	75	73	88	131	75	73	88	131	75	73	88
TWV90802	330RF	xxxxxx	5875	110	129	68	70	84	129	68	70	84	129	68	70	84	129	68	70	84
TWV90802	330RF	xxxxxx	5868	120	125	60	74	79	125	60	74	79	125	60	74	79	125	60	74	79
TWV90802	330RF	xxxxxx	5868	130	121	54	74	49	121	54	74	49	121	54	74	49	121	54	74	49
TWV90802	330RF	xxxxxx	5863	140	114	46	74	32	114	46	74	32	114	46	74	32	114	46	74	32
TWV90802	330RF	xxxxxx	5864	150	107	40	75	20	107	40	75	20	107	40	75	20	107	40	75	20
TWV90801	330LF	xxxxxx	5891	80	138	87	62	3	138	87	62	3	138	87	62	3	138	87	62	3
TWV90801	330LF	xxxxxx	5891	90	134	84	72	0	134	84	72	0	134	84	72	0	134	84	72	0
TWV90801	330LF	xxxxxx	5886	100	131	75	73	88	131	75	73	88	131	75	73	88	131	75	73	88
TWV90801	330LF	xxxxxx	5892	110	129	68	70	84	129	68	70	84	129	68	70	84	129	68	70	84
TWV90801	330LF	xxxxxx	5904	120	125	60	74	79	125	60	74	79	125	60	74	79	125	60	74	79
TWV90801	330LF	xxxxxx	5882	130	121	54	74	49	121	54	74	49	121	54	74	49	121	54	74	49
TWV90801	330LF	xxxxxx	5883	140	114	46	74	32	114	46	74	32	114	46	74	32	114	46	74	32
TWV90801	330LF	xxxxxx	5801	150	107	40	75	20	107	40	75	20	107	40	75	20	107	40	75	20
TWV90704	343RF	xxxxxx	5891	80	138	87	62	3	138	87	62	3	138	87	62	3	138	87	62	3
TWV90704	343RF	xxxxxx	5891	90	134	84	72	0	134	84	72	0	134	84	72	0	134	84	72	0
TWV90704	343RF	xxxxxx	5886	100	131	75	73	88	131	75	73	88	131	75	73	88	131	75	73	88
TWV90704	343RF	xxxxxx	5892	110	129	68	70	84	129	68	70	84	129	68	70	84	129	68	70	84
TWV90704	343RF	xxxxxx	5904	120	125	60	74	79	125	60	74	79	125	60	74	79	125	60	74	79
TWV90704	343RF	xxxxxx	5882	130	121	54	74	49	121	54	74	49	121	54	74	49	121	54	74	49
TWV90704	343RF	xxxxxx	5883	140	114	46	74	32	114	46	74	32	114	46	74	32	114	46	74	32
TWV90704	343RF	xxxxxx	5801	150	107	40	75	20	107	40	75	20	107	40	75	20	107	40	75	20
TWV90703	343LF	xxxxxx	5891	80	138	87	62	3	138	87	62	3	138	87	62	3	138	87	62	3
TWV90703	343LF	xxxxxx	5891	90	134	84	72	0	134	84	72	0	134	84	72	0	134	84	72	0
TWV90703	343LF	xxxxxx	5886	100	131	75	73	88	131	75	73	88	131	75	73	88	131	75	73	88
TWV90703	343LF	xxxxxx	5892	110	129	68	70	84	129	68	70	84	129	68	70	84	129	68	70	84
TWV90703	343LF	xxxxxx	5904	120	125	60	74	79	125	60	74	79	125	60	74	79	125	60	74	79
TWV90703	343LF	xxxxxx	5882	130	121	54	74	49	121	54	74	49	121	54	74	49	121	54	74	49
TWV90703	343LF	xxxxxx	5883	140	114	46	74	32	114	46	74	32	114	46	74	32	114	46	74	32
TWV90703	343LF	xxxxxx	5801	150	107	40	75	20	107	40	75	20	107	40	75	20	107	40	75	20